

Vehicle Routing with Trucks, Drones, and Robots under Dynamically Changing Traffic Conditions

1 Problem Definition

The delivery system is modeled as a directed graph

$$G = (V, \mathcal{A}), \quad V = \mathcal{C} \cup \mathcal{P},$$

where

$$\mathcal{C} = \{1, \dots, n\} \quad \text{denotes customers,} \quad \mathcal{P} = \{0, n+1\} \quad \text{denotes depots.}$$

Each customer $i \in \mathcal{C}$ has demand q_i and a delivery time window $[o_i, e_i]$.

The fleet consists of:

- trucks, indexed by $v \in \mathcal{T}$,
- drones, indexed by $d \in \mathcal{D}$,
- ground delivery robots, indexed by $r \in \mathcal{R}$.

Each truck has capacity Q_t and carries multiple drones and robots. Drones and robots serve at most one customer per sortie and must be launched from and retrieved by a truck.

For each arc $(i, j) \in \mathcal{A}$, we define:

$$d_{ij} \quad (\text{distance}), \quad t_{ij}(\tau) \quad (\text{time-dependent travel time}).$$

Robots operate on a (possibly distinct) pedestrian network $G^{(r)} = (V^{(r)}, \mathcal{A}^{(r)})$, with arc attributes $d_{ij}^{(r)}$ and $t_{ij}^{(r)}(\tau)$.

2 Decision Variables

Truck Routing

$$x_{ij}^v = \begin{cases} 1, & \text{if truck } v \text{ traverses arc } (i, j), \\ 0, & \text{otherwise.} \end{cases}$$

Drone Sorties

$$y_{ijk}^d = \begin{cases} 1, & \text{if drone } d \text{ launches at } i, \\ & \text{serves customer } j, \text{ and rendezvous at } k, \\ 0, & \text{otherwise.} \end{cases}$$

Robot Sorties

$$z_{ijk}^r = \begin{cases} 1, & \text{if robot } r \text{ launches at } i, \\ & \text{serves customer } j, \text{ and rendezvous at } k, \\ 0, & \text{otherwise.} \end{cases}$$

Arrival times at node i are denoted by

$$a_i^{(t)}, \quad a_i^{(d)}, \quad a_i^{(r)}$$

for trucks, drones, and robots, respectively.

3 Constraints

3.1 Customer Service

Each customer is served exactly once:

$$\sum_{v \in \mathcal{T}} \sum_j x_{ji}^v + \sum_{d \in \mathcal{D}} \sum_{j,k} y_{jik}^d + \sum_{r \in \mathcal{R}} \sum_{j,k} z_{jik}^r = 1, \quad \forall i \in \mathcal{C}. \quad (1)$$

3.2 Capacity Constraints

Truck Capacity

$$\sum_{i \in r_v} q_i \leq Q_t, \quad \forall v \in \mathcal{T}. \quad (2)$$

Drone and Robot Capacity

$$q_j \leq Q_d \quad \text{if served by a drone,} \quad q_j \leq Q_r \quad \text{if served by a robot.} \quad (3)$$

3.3 Endurance Constraints

Drone Endurance

$$\sum_{i,j,k} y_{ijk}^d (t_{ij}(\tau) + t_{jk}(\tau)) \leq E_d, \quad \forall d \in \mathcal{D}. \quad (4)$$

Robot Endurance

$$\sum_{i,j,k} z_{ijk}^r (t_{ij}^{(r)}(\tau) + t_{jk}^{(r)}(\tau)) \leq E_r, \quad \forall r \in \mathcal{R}. \quad (5)$$

3.4 Synchronization with Trucks

Drones and robots may only be launched from and retrieved at nodes visited by trucks:

$$y_{ijk}^d + z_{ijk}^r \leq \sum_{v \in \mathcal{T}} x_{ik}^v, \quad \forall i, k. \quad (6)$$

3.5 Time Windows

$$a_i^{(m)} \leq e_i \quad \text{or a penalty is incurred,} \quad m \in \{t, d, r\}. \quad (7)$$

3.6 Route Duration

$$a_{n+1}^{(m)} \leq T_{\max}, \quad m \in \{t, d, r\}. \quad (8)$$

4 Time-Dependent Cost Model

The operational cost of trucks and drones on arc (i, j) is

$$C_{ij}(\tau) = t_{ij}(\tau) C_w + d_{ij} C_{\text{veh}}. \quad (9)$$

The robot travel cost is defined as

$$C_{ij}^{(r)}(\tau) = t_{ij}^{(r)}(\tau) C_w^{(r)} + d_{ij}^{(r)} C_{\text{rob}}. \quad (10)$$

Travel times are obtained from data-driven prediction models reflecting traffic and pedestrian congestion.

5 Objective Function

The total operational cost is

$$\begin{aligned} Z = & \sum_{v \in \mathcal{T}} \sum_{(i,j) \in \mathcal{A}} C_{ij}(\tau) x_{ij}^v \\ & + \sum_{d \in \mathcal{D}} \sum_{i,j,k} (C_{ij}(\tau) + C_{jk}(\tau)) y_{ijk}^d \\ & + \sum_{r \in \mathcal{R}} \sum_{i,j,k} (C_{ij}^{(r)}(\tau) + C_{jk}^{(r)}(\tau)) z_{ijk}^r. \end{aligned} \quad (11)$$

To allow heuristic exploration of infeasible solutions, we minimize the penalized objective

$$\begin{aligned}
P = Z &+ \lambda_E^{(d)} \sum_d \max \left(0, \sum_{i,j,k} y_{ijk}^d (t_{ij} + t_{jk}) - E_d \right) \\
&+ \lambda_E^{(r)} \sum_r \max \left(0, \sum_{i,j,k} z_{ijk}^r (t_{ij}^{(r)} + t_{jk}^{(r)}) - E_r \right) \\
&+ \lambda_Q \sum_v \max \left(0, \sum_{i \in r_v} q_i - Q_t \right) \\
&+ \lambda_T \sum_m \max \left(0, a_{n+1}^{(m)} - T_{\max} \right) \\
&+ \lambda_W \sum_m \sum_{i \in \mathcal{C}} \max \left(0, a_i^{(m)} - e_i \right),
\end{aligned} \tag{12}$$

where $m \in \{t, d, r\}$.